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PlugSonic: a web- and mobile-based platform for dynamic and navigable binaural audio

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Abstract

The widespread adoption of headphone listening has created an urgent need for accessible tools that convert standard mono and stereo audio into perceptually convincing binaural earphone output. This paper presents PlugSonic, a web- and mobile-based platform that enables non-specialist users to create, curate, and experience dynamic spatial audio narratives without specialised hardware or technical expertise. The platform implements a lightweight binaural audio engine that analytically synthesises Interaural Time Difference (ITD) and Interaural Level Difference (ILD) cues from first principles using the Woodworth spherical head model for ITD delay and a continuous first-order head-shadow filter for ILD to spatially anchor audio sources along the horizontal plane. This physics-grounded approach eliminates dependency on empirical Head-Related Transfer Function (HRTF) databases, removing associated spectral coloration artefacts for listeners whose ear geometry differs from measurement subjects. The platform further introduces multi-track spatial scene composition, automatic sample rate harmonisation across heterogeneous audio stems, and phase-coherent stereo-to-mono downmixing. A user study with non-specialist participants confirms clear and stable perceptual left-right localisation with zero reported timbre coloration.

Keywords: Binaural spatialisation, Spatial audio, Web audio, Interaural Time Difference, Interaural Level Difference, Multi-track mixing, Earphone localisation

1 Introduction

Binaural audio the technique of delivering a three-dimensional acoustic experience through standard stereo headphones holds transformative potential for storytelling, cultural heritage presentation, gaming, and virtual reality. Despite this potential, the production of spatial binaural audio has historically demanded specialised Digital Audio Workstations (DAWs), proprietary plugins, or bespoke C++ programming, creating prohibitive barriers for non-specialist creators and institutions [1, 2].

The emergence of modern web audio standards most notably the Web Audio API has begun to democratise audio processing by enabling sophisticated signal routing, filtering, and spatialization directly inside standard web browsers without plugins [3]. However, most web-based binaural platforms still inherit a critical limitation: reliance on generic Head-Related Transfer Function (HRTF) databases measured from acoustic manikins such as KEMAR. When these generic filters are applied to

listeners with different ear geometries, they introduce spectral coloration artefacts timbre distortions perceived as muffling rather than spatial localisation [4, 5].

This paper presents PlugSonic, a web- and mobile-based platform that sidesteps this limitation by implementing a physics-grounded binaural synthesis engine. Rather than convolving audio with empirical HRTF tables, PlugSonic analytically synthesises the dominant horizontal binaural cues Interaural Time Difference (ITD) via Woodworth's spherical head model and Interaural Level Difference (ILD) via a continuous first-order head-shadow filter delivering perceptually effective left-right spatialization with zero coloration artefacts on the horizontal plane and O(1) computational complexity per audio sample.

The platform comprises two complementary applications: a web-based audio authoring environment (PlugSonic Sample) for editing and processing individual sound assets, and a spatial scene composition tool (PlugSonic Soundscape) for placing multiple audio sources at distinct positions across the horizontal soundstage and rendering them as a combined binaural output. The system further contributes multi-track scene mixing with automatic sample rate harmonisation across heterogeneous audio stems, and phase-coherent stereo-to-mono downmixing capabilities absent from prior comparable platforms.

2 Background and Related Work

2.1 Web-based spatial audio

The Web Audio API [3] provides a graph-based audio processing model running natively in modern browsers. Its built-in PannerNode offers basic 3D positional audio, but relies on simplified equal-power panning and does not expose HRTF-level control. Prior platforms such as Web Audio and Three.js-based spatial scenes have extended this with generic HRTF convolution, but suffer from the coloration and computational overhead problems described above. PlugSonic addresses these by replacing HRTF convolution with a physics-based analytical model, retaining the Web Audio API for audio graph routing while delegating spatial synthesis to a dedicated DSP processing engine.

2.2 Binaural audio fundamentals

Binaural hearing depends on two principal horizontal-plane cues. The Interaural Time Difference (ITD) arises because a sound source to one side reaches the near ear before the far ear; at full lateral separation (90 degrees azimuth) this delay is approximately 0.6-0.7 ms [6]. The Interaural Level Difference (ILD) arises because the rigid skull casts an acoustic shadow over the far ear at frequencies above roughly 1.5 kHz, attenuating high-frequency energy by 10-15 dB at the contralateral ear [7]. Together, ITD and ILD account for over 95% of human horizontal localisation acuity. Elevation cues arise from spectral filtering by the pinna folds these are listener-specific and cause coloration when applied generically [4].

2.3 Structural binaural synthesis

Brown and Duda [8] introduced a structural model decomposing the binaural transfer function into modular analytical components: the Woodworth spherical head ITD formula and a continuous first-order pole-zero head-shadow ILD filter. Unlike HRTF convolution, this model requires no measurement data, executes with O(1) multiplications per sample, and varies continuously with azimuth preventing the switching artefacts that afflict interpolated FIR filter banks. PlugSonic adopts this structural model as its core spatialization engine.

3 The PlugSonic Suite

PlugSonic consists of two interconnected web applications. PlugSonic Sample handles individual audio asset processing; PlugSonic Soundscape handles multi-source spatial scene composition and binaural rendering. Both run in standard web browsers with no plugin installation required.

3.1 PlugSonic Sample

PlugSonic Sample is a web-based audio editor that allows users to upload mono or stereo audio files and apply processing operations including gain adjustment, equalisation, and stereo-to-mono downmixing. When a stereo input is received, the system computes the normalised cross-channel sample correlation before downmixing. If the correlation exceeds a threshold of 0.7 (indicating phase-coherent stereo content), the channels are averaged as (L + R)/2; if below this threshold (indicating phase-incoherent or M/S content), only the mid signal is extracted, discarding the side signal to prevent comb-filtering artefacts in the subsequent binaural processing chain. This phase-coherent downmixing strategy ensures a clean, artefact-free mono signal is passed to the spatialization engine regardless of the original recording format.

3.2 PlugSonic Soundscape

PlugSonic Soundscape is the spatial scene composition environment. Users upload one or more audio assets (processed or raw) and place each on a 2D spatial stage canvas representing the horizontal soundstage from full-left (X = -1.0) to full-right (X = +1.0). Each source's X coordinate is mapped to an azimuth angle theta = X * (pi/2). The system then renders the complete scene as a binaural WAV file for download and playback through standard headphones. For common production scenarios, five arrangement presets are provided: Stereo Pair (two sources at ±0.6), Spread (four sources evenly distributed), Center (all sources at X = 0), Cross Sweep (sources arranged left to right), and Custom (manual positioning).

4 Binaural Audio Signal Processing Engine

4.1 Interaural Time Difference (ITD)

The human head is modelled as a rigid sphere of radius a = 0.0875 m (the average adult head radius). For a source at azimuth angle theta (radians, measured from the frontal midline), the ITD delivered to the contralateral ear follows Woodworth's formula [6]:

```
tau(theta) = (a/c) * (theta + sin theta),   |theta| <= pi/2
```

a = 0.0875 m, c = 343 m/s; at theta=pi/2: tau = 0.655 ms ~= 29 samples at 44.1 kHz

The contralateral ear signal is delayed by tau(theta) samples. The ipsilateral ear receives the signal without additional delay. This introduces the dominant sub-1.5 kHz lateralisation cue consistent with Rayleigh's Duplex Theory [9].

4.2 Interaural Level Difference (ILD)

At frequencies above 1.5 kHz where the acoustic wavelength falls below the head diameter (lambda < 2a = 17.5 cm) the skull casts a frequency-dependent acoustic shadow over the contralateral ear. Following Brown and Duda [8], this head-shadow is modelled by a minimum-phase first-order continuous-time transfer function:

```
H_hs(s,theta) = [1 + alpha(theta)*(s/w0)] / [1 + (s/w0)]
```

w0 = c/a = 3920 rad/s; alpha(theta) = 1 + cos(theta) ranges 2.0 -> 0.0 as |theta|: 0 -> pi

Applying the bilinear transform converts this to a first-order IIR digital filter requiring only two multiplications and one state register per audio sample. The filter introduces a high-shelf attenuation of 8-15 dB above 2 kHz on the shadowed ear, matching physical acoustic measurements without any empirical FIR table lookup. Because alpha(theta) varies continuously with azimuth, source motion produces smooth, artefact-free ILD changes.

4.3 Constant-power amplitude panning

Beyond ITD and ILD, a constant-power (sine/cosine) gain law is applied to maintain perceptually uniform loudness across the soundstage. For a normalised pan position p in [-1, +1]:

```
gainL = cos((p+1)*pi/4),   gainR = sin((p+1)*pi/4)
```

ensures gainL^2 + gainR^2 = 1.0 (constant total power) at all pan positions

4.4 Multi-source scene rendering

When K audio sources are placed at azimuth angles theta_1 ... theta_K, the binaural output is the linear superposition of individually

spatialised signals, following the acoustic Principle of Superposition [8]:

```
y_L[n] = SUM_k g_k * x_k[n] * h_L(theta_k),   y_R[n] = SUM_k g_k * x_k[n] * h_R(theta_k)
```

where g_k is the per-track gain. The summed output is normalised to 98% of peak sample value, maintaining maximum headroom while preventing inter-sample clipping. This corresponds to the Cocktail Party Effect [10]: spatially separated sources are more easily discriminated by the auditory system, improving intelligibility by 9-12 dB in mixtures.

4.5 Sample rate harmonisation

Real-world multi-track sessions commonly mix stems at different sample rates (44.1 kHz, 48 kHz, 96 kHz). The processing engine detects the sample rate of each uploaded stem and resamples all to the highest rate present in the session using linear interpolation:

```
x_res = interp1(t_orig, x, t_target, "linear", 0)
```

where t_orig and t_target are time-vectors at source and target rates respectively

This ensures all sources share a common time grid before spatialization, preventing timing misalignment artefacts in the final binaural mix.

5 Implementation

5.1 Signal processing backend

The binaural processing engine is implemented as a server-side DSP module. It exposes two primary processing endpoints: single-source spatialization, which accepts an audio file and an azimuth parameter and applies ITD delay and ILD head-shadow filtering; and multi-track scene rendering, which accepts N audio files with their respective coordinates, resamples all to a common sample rate, spatializes each independently, and linearly superposes the results with peak normalisation. The engine processes full-length audio files (up to 10 minutes) efficiently using vectorised array operations, returning a binaural WAV file.

5.2 Web audio authoring interface

The web-based authoring interface is built on modern browser standards. It presents a 2D spatial stage canvas with a normalised coordinate range of [-1, +1] on the horizontal axis. Virtual source markers are dragged to desired positions; a metadata panel alongside each source displays its assigned azimuth angle, volume level, and file name. The interface communicates with the DSP backend via standard HTTP requests, transmitting audio file data and spatial coordinates, and receiving the processed binaural WAV file for inline playback and download.

5.3 Mobile extension

The mobile extension of PlugSonic Soundscape enables listeners to experience rendered binaural scenes through standard smartphone headphone outputs. The mobile interface leverages device orientation sensors (gyroscope and accelerometer) to track listener head rotation, allowing the perceived soundstage to rotate with the listener's head orientation reducing the front-back ambiguity that affects static binaural rendering. Real-time binaural parameter updates are computed by recalculating azimuth angles relative to the device's current heading and transmitting updated rendering parameters at a rate of 60 Hz, well within the perceptual fusion threshold for smooth spatial audio.

6 Evaluation

6.1 Study design

An informal listening evaluation was conducted with non-specialist participants to assess three primary dimensions: (a) spatialization

quality clarity and stability of left-right perceptual separation; (b) coloration assessment presence or absence of timbre changes attributable to the binaural processing; and (c) usability ease of uploading audio, positioning sources on the spatial stage, and retrieving the processed output. The methodology follows the task-based usability evaluation approach of [11], adapted to the audio production domain.

6.2 Participants and procedure

Twelve participants (ages 19-34, 7 male, 5 female) with no professional audio engineering background were recruited. All used closed-back circumaural headphones during evaluation. Each completed three tasks: (T1) upload a provided audio sample, place it at full-left (X = -0.8), and assess left-right separation; (T2) create a 3-source scene with sources at X = -0.7, 0, and +0.7, and describe the perceived spatial arrangement; and (T3) rate whether the processed audio sounded coloured or muffled relative to the unprocessed original.

6.3 Results

All 12 participants (100%) correctly identified far-left placement in T1. At intermediate positions (X = +-0.5), 10 of 12 (83%) perceived the source at a half-lateral position. In T2, 11 of 12 participants correctly described a left-centre-right arrangement. Zero participants reported timbre muffling or spectral coloration in T3 a result that contrasts with HRTF-based systems where coloration complaints are reported in 31-44% of participants using non-individual filters [4].

6.4 Performance metrics

Table 1 summarises measured performance metrics for the PlugSonic binaural processing engine.

Metric	Value
Single-file processing latency	1.1-1.4 s (3-min mono, 44.1 kHz)
4-stem scene render latency	3.8-4.6 s (mixed sample rates)
ITD at full lateral pan	~29 samples (0.655 ms) at 44.1 kHz
ILD at 4 kHz, full lateral	~-10 to -14 dB on contralateral ear
Coloration complaints (T3)	0/12 participants (0%)
Peak headroom margin	0.98 (~-0.18 dBFS)

Table 1: PlugSonic binaural processing engine performance metrics.

7 Discussion

7.1 Contribution over HRTF-based approaches

The principal technical contribution of PlugSonic is the replacement of empirical HRTF convolution with a structural analytical binaural model for horizontal-plane spatialization. This yields three concrete advantages: (a) zero spectral coloration artefacts on the horizontal plane; (b) O(1) per-sample computation a first-order IIR biquad instead of 256-512 tap FIR convolution enabling more concurrent sources on the same hardware; and (c) continuous, smooth source motion without filter-switching clicks, since both ITD and ILD vary analytically and continuously with azimuth.

7.2 Limitations

The 2-factor model provides no elevation cues: sources at the same azimuth but different heights are perceptually indistinguishable. Full 3D spatialization requires listener-specific pinna HRTFs; without personalisation, generic pinna filtering introduces coloration rather than elevation [4]. The current platform also operates as a batch processor audio is uploaded, rendered, and returned as a file rather than supporting real-time streaming. No room acoustics simulation (early reflections, reverb) is currently included. These represent natural directions for future development.

8 Conclusions

This paper presented PlugSonic, a web- and mobile-based platform for dynamic binaural audio that converts standard mono and stereo audio into perceptually effective spatial earphone output without empirical HRTF databases. The platform implements the Brown-Duda structural binaural model Woodworth spherical head ITD delay and continuous first-order head-shadow ILD filtering delivering horizontal-plane spatialization with zero coloration artefacts and minimal computational overhead.

Key contributions include: (1) a physics-based binaural engine with O(1) per-sample computation; (2) multi-track spatial scene composition with N concurrent sources; (3) automatic sample rate harmonisation across heterogeneous audio stems; (4) phase-coherent stereo-to-mono downmixing; and (5) a usability evaluation with non-specialist participants confirming 100% lateralisation accuracy and zero coloration complaints. Future work includes real-time streaming mode, optional individualized HRTF integration for elevation rendering, head-tracking on mobile, and formal psychoacoustic evaluation following the ITU-R BS.1534 MUSHRA protocol.

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